3.3 Pre-Test: Trigonometric Functions — Markscheme

1. [Maximum mark: 9]

(a) $135^\circ = \dfrac{3\pi}{4}$ **A1**

(b) $2.4 \times \dfrac{180}{\pi}$ **M1**

$= 138^\circ \quad (137.509\ldots)$ **A1**

A1 for 138 or the display with an ellipsis; 137.5 with no ellipsis earns A0 (markscheme-spec §21). Wrong method: $2.4 \times \frac{\pi}{180} = 0.0419$ *earns M0A0.*

Sector: $r = 9$, $\theta = 1.2$.

(c) $l = 9 \times 1.2$ **M1**

$= 10.8$ cm **A1**

(d) $A = \dfrac{1}{2} \times 9^2 \times 1.2$ **M1**

$= 48.6$ cm^2 **A1**

Wrong methods: the degree formula, $\frac{1.2}{360} \times 2\pi \times 9 = 0.188$ *in (c) or* $\frac{1.2}{360} \times \pi \times 9^2$ *in (d), earns M0A0; r for* r^2, $\frac{1}{2} \times 9 \times 1.2 = 5.4$, *earns M0A0.*

(e) $60 = \dfrac{1}{2} \times 9^2 \times \theta$ **M1**

$\theta = 1.48 \quad (1.48148\ldots)$ **A1**

Accept the exact value $\frac{40}{27}$. *Wrong methods: the* $\frac{1}{2}$ *dropped,* $\frac{60}{81} = 0.741$, *earns M0A0; the arc-length formula,* $\frac{60}{9} = 6.67$, *earns M0A0. An answer in degrees only,* 84.9°, *earns M1A0.*

2. [Maximum mark: 10]

(a)(i) $\cos\dfrac{3\pi}{4} = -\dfrac{\sqrt{2}}{2}$ **A1**

(a)(ii) $\sin\dfrac{5\pi}{3} = -\dfrac{\sqrt{3}}{2}$ **A1**

(a)(iii) $\tan\dfrac{4\pi}{3} = \sqrt{3}$ **A1**

Accept $-\frac{1}{\sqrt{2}}$ *and* $\frac{3}{\sqrt{3}}$. *Decimals from the GDC (*-0.707, -0.866, 1.73*) earn A0: exact values are asked for. Common errors, each A0: wrong sign in (ii) or (iii).*

(b)(i) $\sin^2\theta = 1 - \left(-\dfrac{5}{13}\right)^2 = \dfrac{144}{169}$ **M1**

$\sin\theta = \pm\dfrac{12}{13}$ **A1**

$\sin\theta = -\dfrac{12}{13}$, since θ is in the third quadrant **A1**

(b)(ii) $\tan\theta = \dfrac{-12/13}{-5/13} = \dfrac{12}{5}$ **A1**

Accept a right-triangle method (sides 5, 12, 13) for the M1A1. The final A1 in (i) is for the negative sign. Wrong method: the GDC route $\sin\left(\arccos\left(-\frac{5}{13}\right)\right) = +\frac{12}{13}$ *gives the second-quadrant angle, earning M1A1A0, and then* $\tan = -\frac{12}{5}$*, A0 unless FT. FT (ii) from the candidate's* $\sin\theta$*. Decimals (*-0.923, 2.4*) earn A0 for exactness, except that* 2.4 *is exact: accept* 2.4 *in (ii).*

(c) $\sin x = -\dfrac{1}{2}$ **M1**

$x = \dfrac{7\pi}{6},\ \ x = \dfrac{11\pi}{6}$ **A1A1**

Wrong methods: the GDC value $x = -\frac{\pi}{6}$ *(*-0.524*) is outside the interval, A0;* $x = \frac{\pi}{6}, \frac{5\pi}{6}$ *(sign of* $\sin x$ *wrong) earns M0A0A0. Decimals (*3.67, 5.76*) earn A1A0 at most: the question asks for multiples of* π*. Extra values outside the interval earn the final A0.*

3. [Maximum mark: 10]

$f(x) = a\cos(bx) + d$, maximum $(0,\ 7)$, minimum $\left(\frac{\pi}{6},\ -1\right)$.

(a) period $= 2 \times \dfrac{\pi}{6} = \dfrac{\pi}{3}$ **A1**

(b) $a = \dfrac{7-(-1)}{2} = 4$ **A1**

$d = \dfrac{7+(-1)}{2} = 3$ **A1**

$b = \dfrac{2\pi}{\pi/3} = 6$ **A1**

FT b from the candidate's period in (a). Wrong methods, each A0: $b = \frac{\pi}{3}$ (period used as b); $a = 8$ (max − min not halved); $d = 4$.

(c) $h(x) = \dfrac{1}{2}(4\cos(6x) + 3) - 5$ **M1**

$= 2\cos(6x) + 1.5 - 5$ **A1**

$h(x) = 2\cos(6x) - 3.5$ **A1**

M1 for the stretch applied to the whole of f and the translation after it. Accept $-\frac{7}{2}$. Wrong methods: stretching only the cosine term, $2\cos(6x)+3-5 = 2\cos(6x)-2$, earns M0A0A0; translating before stretching, $\frac{1}{2}(f(x)-5) = 2\cos(6x) - 1$, earns M0A0A0. FT from the candidate's a, b, d.

(d) Vertical stretch, scale factor 4 **A1**

Horizontal stretch, scale factor $\dfrac{1}{6}$ **A1**

Vertical translation 3 up **A1**

"Describe fully" needs words (markscheme-spec §19): one mark per transformation, with its type, direction and size. Accept "horizontal compression by a factor of 6". Accept "translation by $\binom{0}{3}$". Reject "move". A correct chain of equations with no words earns at most 1 mark. Order: the vertical stretch must come before the vertical translation; the horizontal stretch may come anywhere. Deduct 1 mark for a wrong order. Wrong method: horizontal stretch with scale factor 6, A0. FT from the candidate's values in (b).

4. [Maximum mark: 11]

$D(t) = 2.5\sin(0.5t) + 6$, t hours after midnight.

(a) maximum = 8.5 m **A1**

minimum = 3.5 m **A1**

(b) period $= \dfrac{2\pi}{0.5}$ **M1**

$= 4\pi = 12.6$ hours $\quad (12.566\ldots)$ **A1**

Accept the exact value 4π*. Wrong method:* $2\pi \times 0.5 = \pi$ *earns M0A0.*

(c) $D(9) = 2.5\sin(0.5 \times 9) + 6$ **M1**

$= 3.56$ m $\quad (3.55617\ldots)$ **A1**

Wrong method: degree mode gives 6.20 *(6.19614...), earning M1A0.*

(d) $2.5\sin(0.5t) + 6 = 7$ **M1**

GDC: graph $y = D(t)$ *and* $y = 7$ *for* $0 \leq t \leq 12$*; find both intersections.*

$t = 0.823$ $(0.823033\ldots)$ and $t = 5.46$ $(5.46015\ldots)$ **A1**

length $= 5.46015\ldots - 0.823033\ldots = 4.64$ hours $\quad (4.63711\ldots)$ **A1**

M1 for the equation or inequality $D(t) \geq 7$*, or a sketch with the line* $D = 7$*. The first A1 needs both values. Accept 4.64 from the rounded values. Wrong methods: one intersection given as the answer earns M1A0A0; degree mode gives no intersection in* $0 \leq t \leq 12$*, M1A0A0.*

(e) $\sin(0.5t) = -1$, so $0.5t = \dfrac{3\pi}{2}$ **M1**

$t = 3\pi = 9.42$ hours after midnight $\quad (9.42477\ldots)$, about 09:25 **A1**

Accept 3π*, 9.42 or 09:25. Accept the GDC minimum of the graph for the M1. Wrong methods:* $0.5t = \frac{\pi}{2}$ *gives* $t = \pi$ *(the first maximum), M0A0;* $t = 12.6$ *(one period), M0A0.*